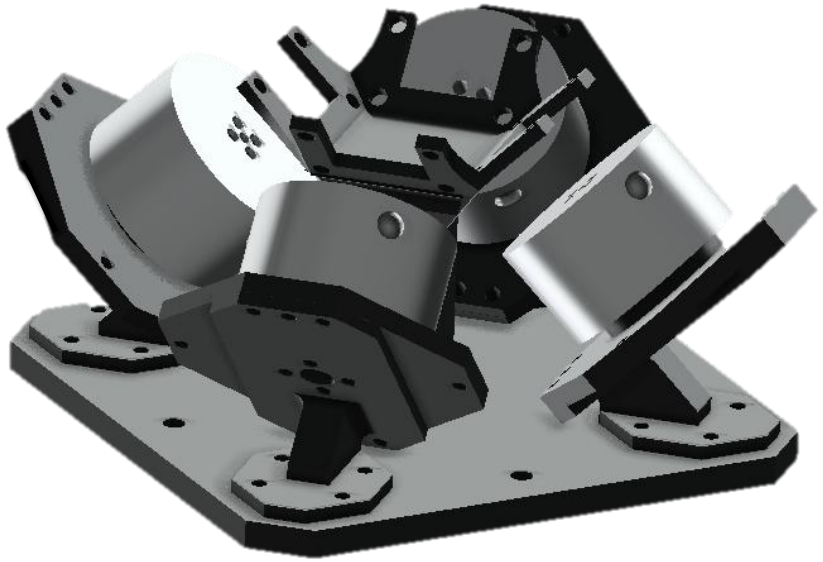
Redundant CubeSat Attitude Stabilisation Module (RCAS)



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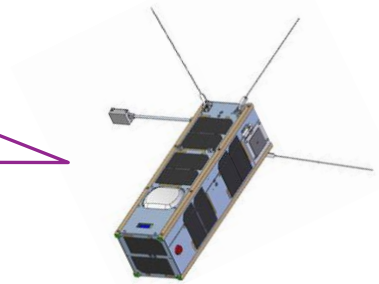
Product Overview

RCAS is a fault-tolerant attitude control module for CubeSats. It uses reaction wheels and orientation sensors to control the attitude of CubeSats.

It allows very small satellites to point at and track moving objects, such as a point on earth or at distant stars, and continue operating even if a key component fails.

This aims to address a major issue in spaceflight: the failure of a single mechanical component can cause a complete loss of control.

Reaction wheels are small, motor-driven wheels, used to rotate a satellite in the opposite direction they spin. Allowing precise orientation without using fuel.



Product Showcase

The design shown uses four reaction wheels, instead of the minimum three required to turn a satellite. The fourth wheel provide full 3-axes control even after a wheel failure, preventing missions from ending prematurely.

RCAS reduces the risk of complete mission failure while remaining affordable and accessible.

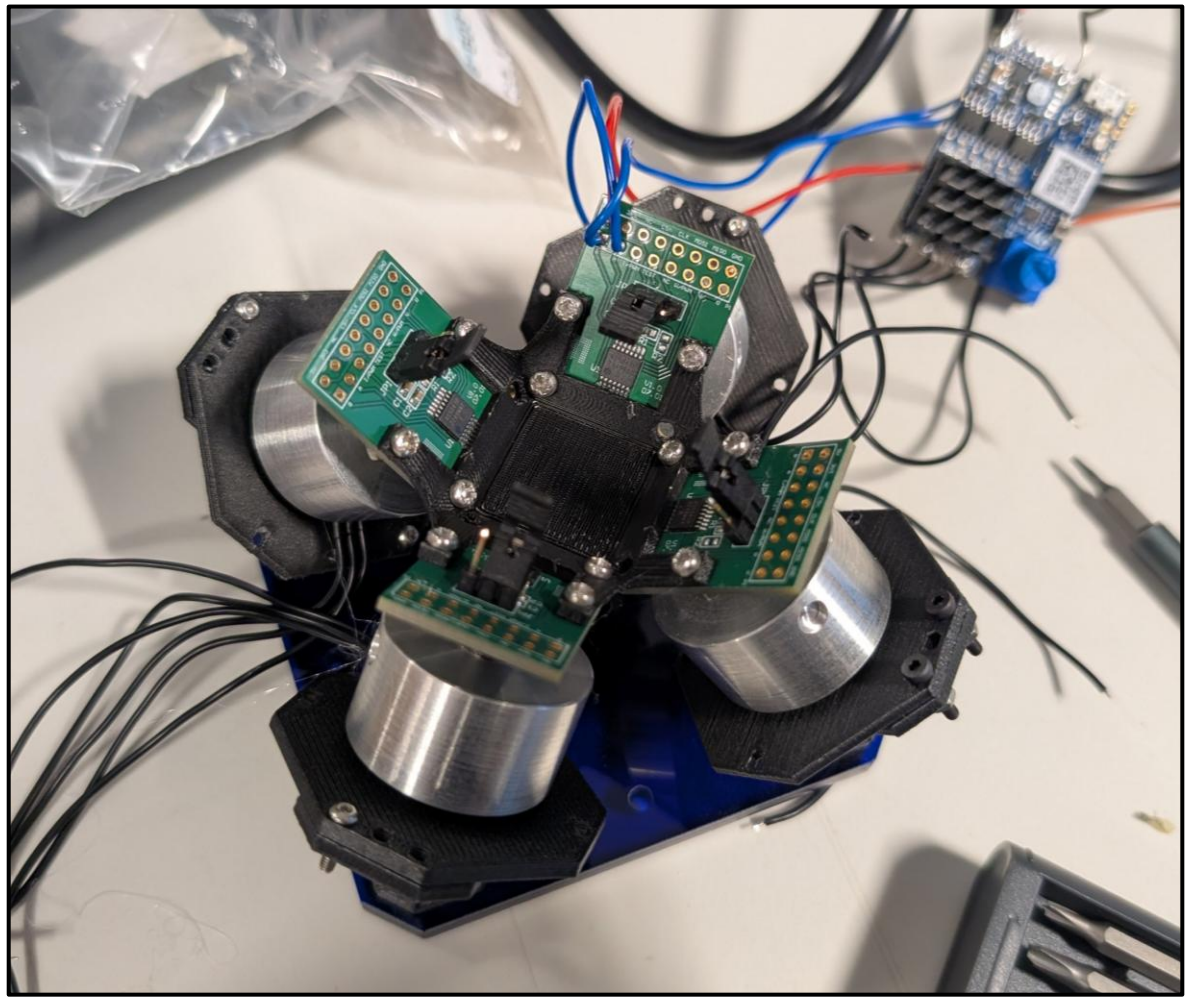
Preliminary simulation: tracking a point in zero gravity (Simulink)

Current Development

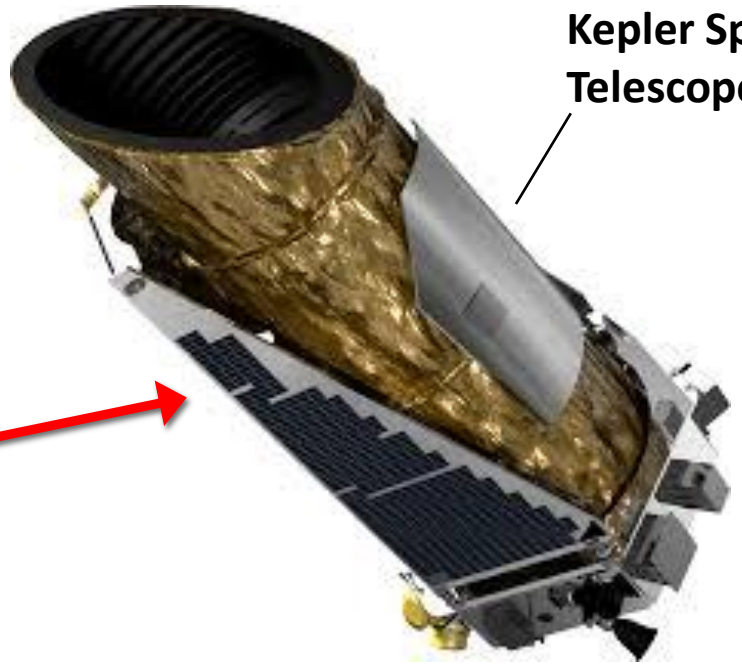
Currently, a full simulated model of the design is complete.

A hardware protype has been manufactured and has tested sensor data and motor control.

A full fault detection and isolation algorithm needs to be implemented and tested on physical hardware.



Why is redundancy so important?



Kepler Space Telescope

Also used reaction wheels

The Kepler Spacecraft Failure

In 2012, the Kepler Telescope Spacecraft lost functionality of one of its four reaction wheels, used to point the telescope at distant planets and stars.

The redundant design of its reaction wheel array allowed it to complete its primary mission

The failure of a second wheel a year later, crippled the pointing capabilities of the spacecraft and limited the scope of future missions

[1]
"NASA Ends Attempts to Fully Recover Kepler Spacecraft, Potential New Missions Considered - NASA," NASA, Aug. 15, 2013. https://www.nasa.gov/news-release/nasa-ends-attempts-to-fully-recover-kepler-spacecraft-potential-new-missions-considered/

Target Market

- Research groups and academic institutions
- Satellite companies, small-scale aerospace contractors
- National space administrations of small or developing nations
- Private and commercial ventures outside of government and defence

Key Stakeholders

Users

- Research/Universities
- Space start-ups

Investors

- Crowdfunding Investors
- Space focused investors or grants

Regulators

- National space agencies
- Launch providers

Suppliers

- Components suppliers and manufacturers

Indirect

- Educational institutions: Adoption and growth of the ecosystem
- Wider space community: Widening access to spacecraft technology
- Future investors: Government grants and commercial investment from large companies

Unique Selling Points

A reliability-first design at CubeSat scale

Open documentation for design architecture

High performance to cost ratio

Unique Selling Points

The compact design offers reliable CubeSat control, even with the complete failure of a reaction wheel.

The module has been designed to be low-cost, featuring commercial components and uses additive manufacturing (3D printing) to reduce manufacturing cost and production time.

A transparent architecture makes it very attractive for small institutions, as most components in the aerospace market are proprietary and difficult to integrate.

Competition

Company	Device	Cost (USD)	Actuators	Failure tolerance
CubeSpace	Closed Loop ADCS, 3-axis small	$34,000	x3 reaction wheels	No
Tensor Tech	CMG-10m	$22,000	x1 control-moment gyro	No
Space Resources Laboratory	Integrated ADCS + Reaction Wheels	$15,000 + $28,000	X4 reaction wheels	Yes

Costs

Materials:
Electrical components = £235
Filament, aluminium = £45
Mechanical components = £40
Total material cost = £320

Labour:
Machinist
Assembly technician
Engineer

Tools:
3D Printer
Lathe
Multimeter
Power supply
Computer and software licenses

Sale price:
Materials = £320
Production + Overheads ≈ £760
Profit margin of 40%

Sales price per unit = £1512

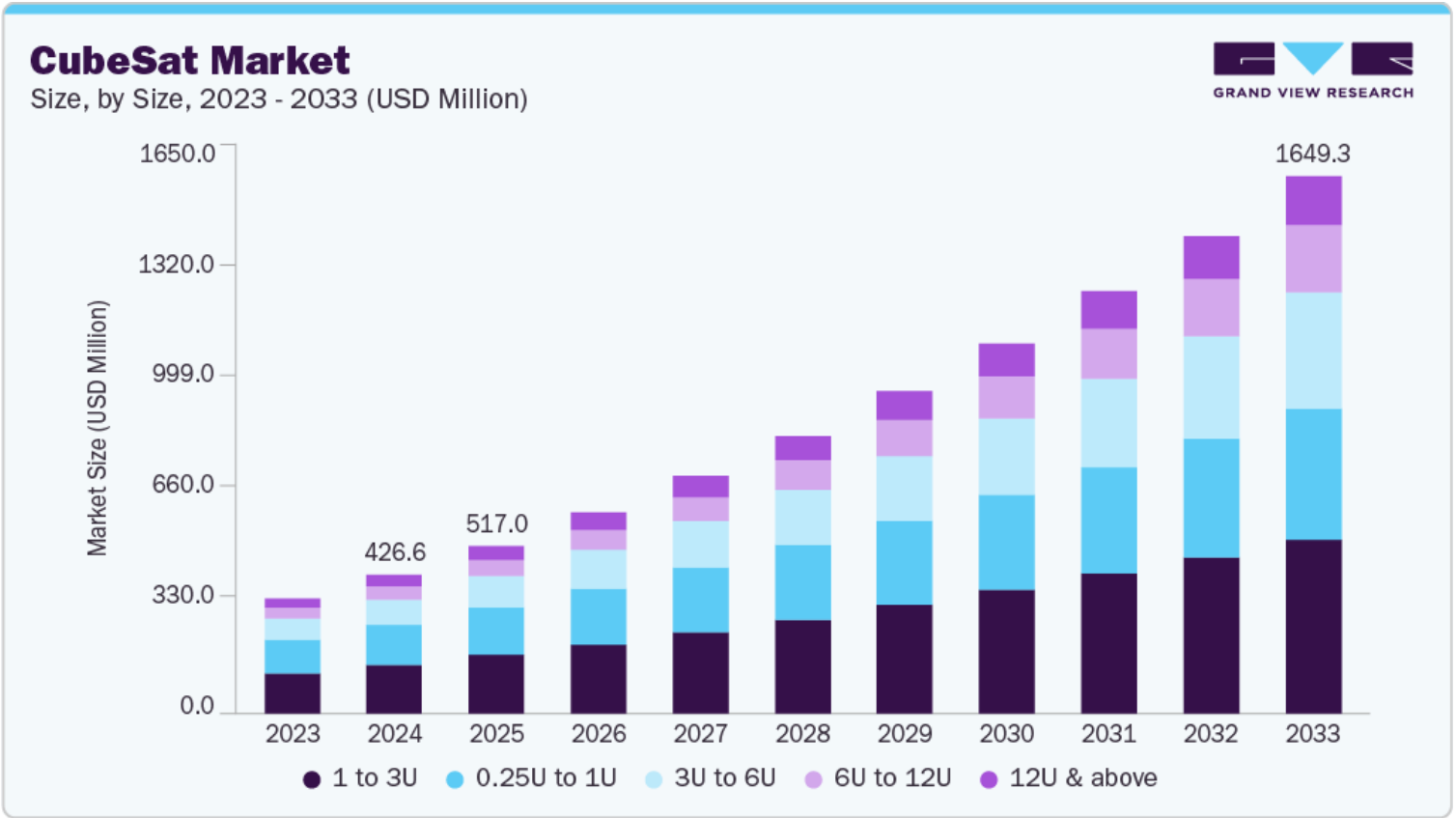
This is over 10x cheaper than the nearest market competitors!

Additional costs may occur as the product is developed but would still fall well short of the most basic models on the market.

Market Opportunity

According to Grand View Research, the CubeSat market currently sits at over $400 million and is projected to exceed $1.5 billion by 2033.

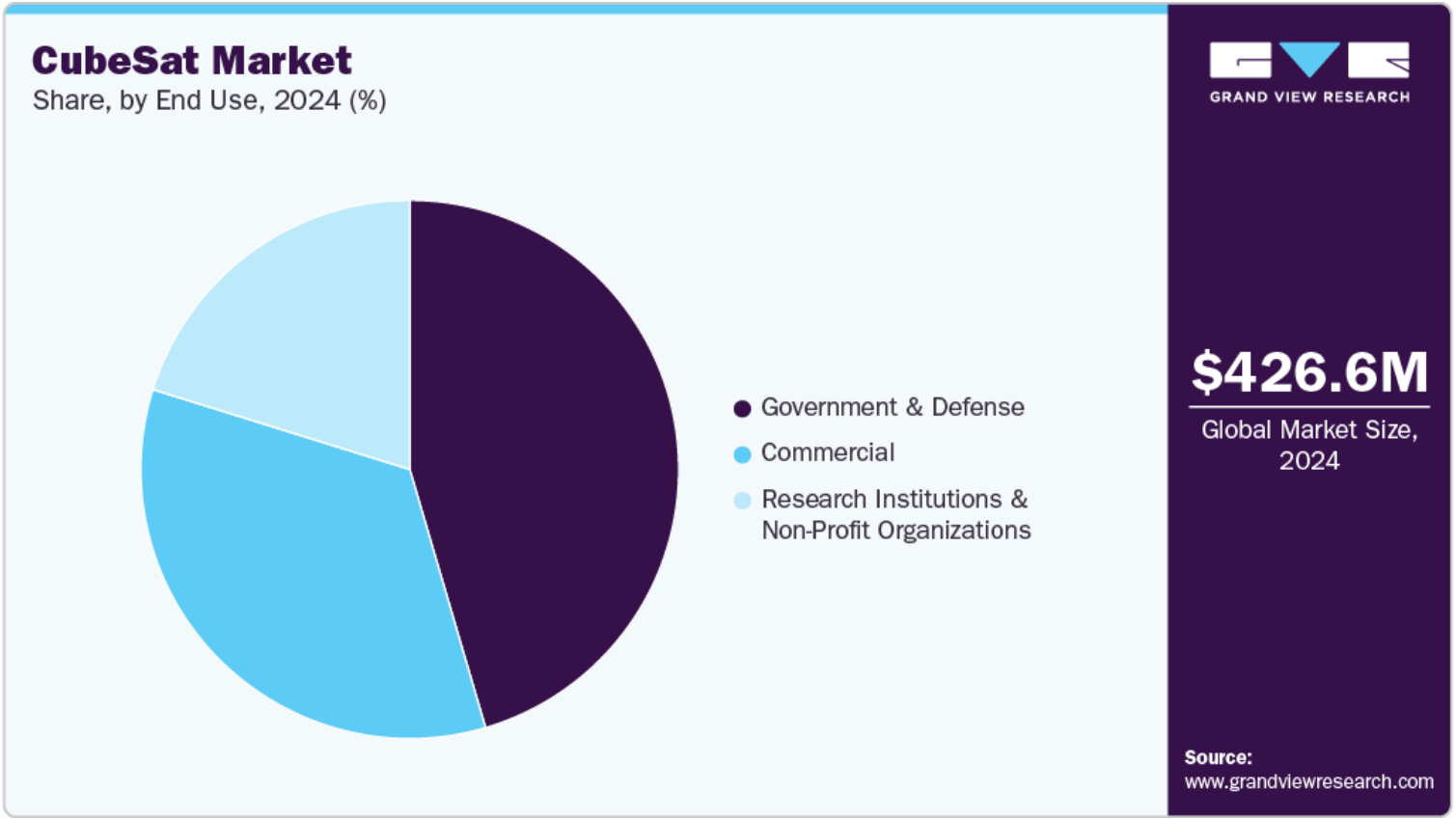
This is driven by technological advances in satellite technology and lower deployment costs offered by reusable launch services (such as SpaceX).



Market Opportunity

- Most of the CubeSat market is comprised of commercial and research institutions.

- These groups lack the resources to acquire and integrate commercial satellite components and often rely on in-house manufacturing.



[1]
"CubeSat Market Size, Share & Trends | Industry Report, 2033," Grandviewresearch.com, 2024. https://www.grandviewresearch.com/industry-analysis/cubesat-market-

Implementation

Stage 1: Research and Development

- Develop a cost-optimised and professional design
- Research endurance for space missions

Stage 2: Testing

- Test the module under various degradation and failure scenarios
- Ensure the design is reliable and mission ready

Stage 3: Analysis

- Evaluate the performance of the design (sensors, motors…)
- Analyse improvements for final iteration

Stage 4: Final Design and Production

- Implement the design using space-ready hardware
- Test and debug produced units

Stage 5: Documentation and Support

- Fully document the hardware and software
- Provide continuous community support to ensure longevity

SWOT Analysis

Strengths

- Low-cost module compared to current market options
- Fault-tolerant design
- Transparent architecture
- Strong alignment with research

Weaknesses

- Unproven product compared to established vendors
- Small development team
- Limited initial production capacity
- Requires user technical knowledge for integration

Opportunities

- Rapid growth in university and commercial missions
- Large gap between commercial products and custom solutions
- Community driven development through open-source and crowd funding

Threats

- Established competitors lowering prices
- Component supply chain disruptions
- Regulation changes/ compliance issues
- Launch delays reducing mission demand

Sustainability, Ethics and Environmental Impact

Environmental Impact

- Compact design minimises launch mass and reduces energy usage compared to standard larger commercial systems
- Lower mission failure rate reduces the risk of uncontrollable decommissioned satellites
- Aligned with IADC Space Debris Mitigation Guidelines

Ethical and Social Impact

- Increased accessibility for small teams, universities or developing nations to access spacecraft design and deployment
- Supports education through transparent design of real space hardware
- Open documentation provides accountability and reduces the risks of hidden failure modes or unsafe operation

Economic Impact

- Lower cost per mission makes the overall cost more viable for smaller institutions
- Enables new CubeSat companies and start-ups by reducing a big technical barrier in satellite development
- Introduces competition in a market dominated by high-cost suppliers, forcing vendors to reduce prices or improve designs

Future Improvements

Addition of advanced sensors: Star tracker, sun sensor	→	Enhanced attitude determination
Magnetorquers for active detumbling	→	Allows for coarse control
Support for different configurations	→	Better compatibility with different satellites

Conclusion

The RCAS system provides accessible, low-cost satellite technology to organisations that would otherwise lack the resources to take advantage of space applications.

Thank you for listening to my presentation